

Операционные системы

Командная оболочка Midnight Commander

Ирсана Атабаева

2026-03-23

1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
3. Выводы по проделанной работе

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

1.2 Задачи лабораторной работы

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. 2. Процесс выполнения лабораторной работы

2.1 Работа с Midnight Commander

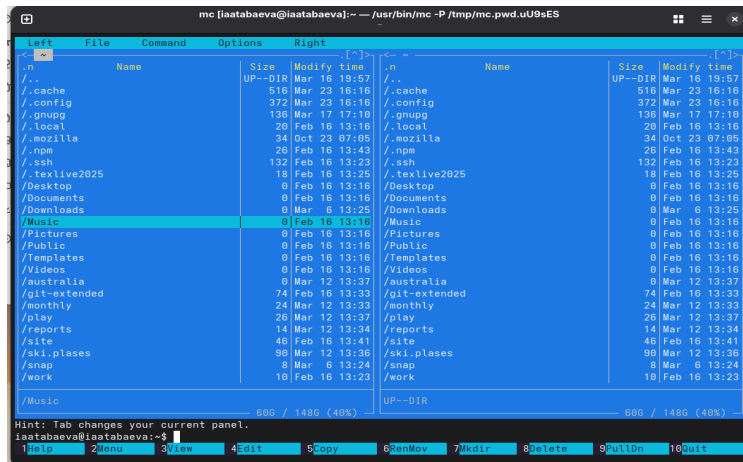


Рисунок 1: Запуск mc

2.2 Работа с Midnight Commander

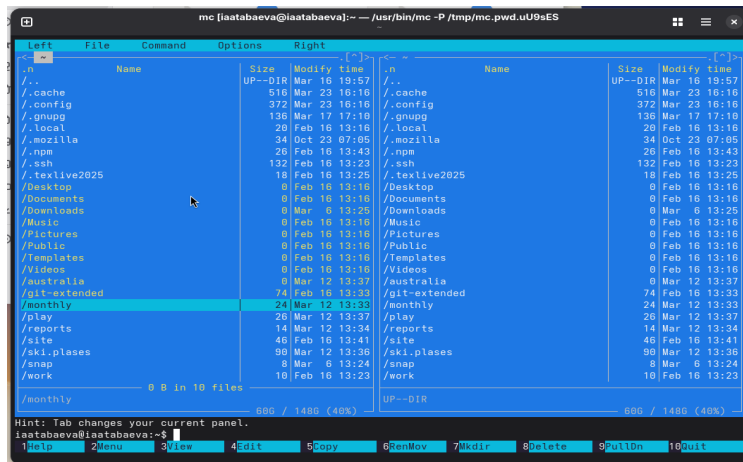


Рисунок 2: Выделение

2.3 Работа с Midnight Commander

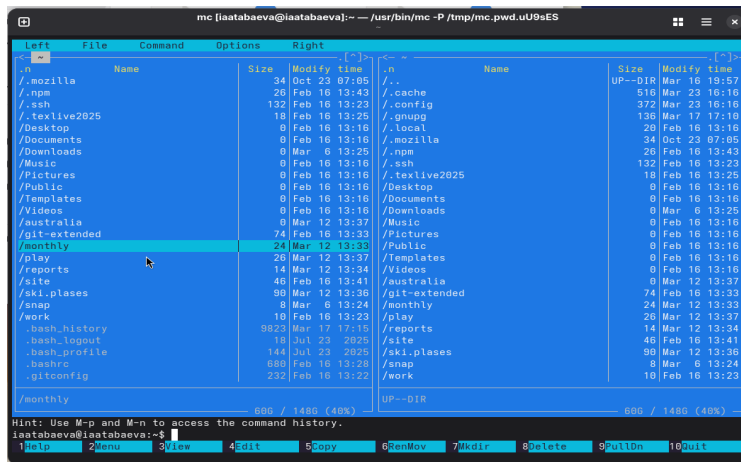


Рисунок 3: Отмена

2.4 Работа с Midnight Commander

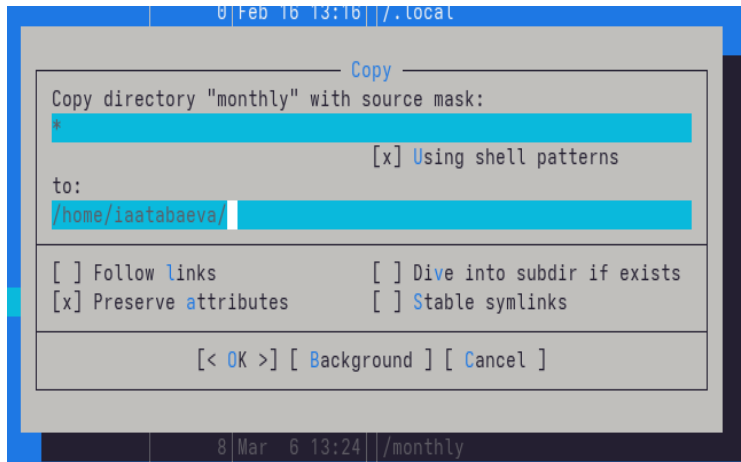


Рисунок 4: Копирование

2.5 Работа с Midnight Commander

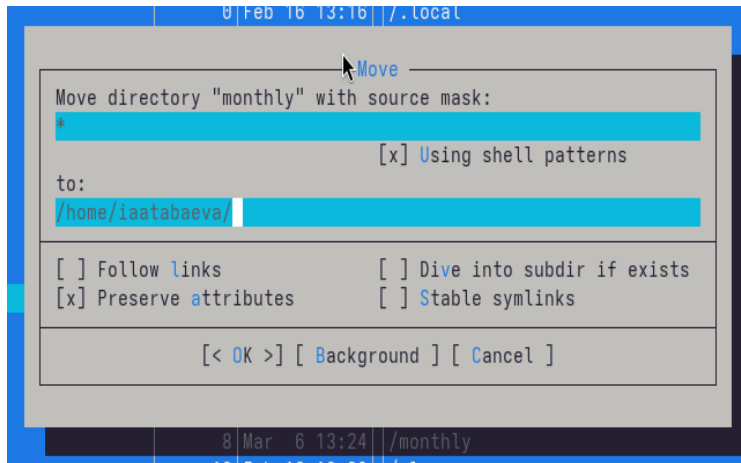


Рисунок 5: Перемещение

2.6 Работа с Midnight Commander

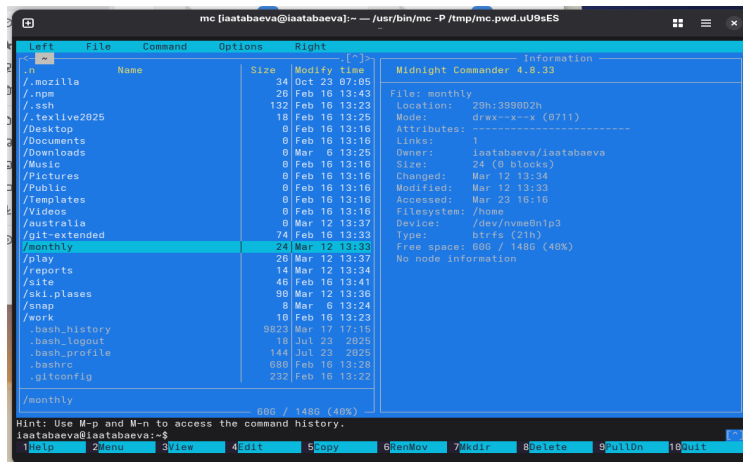


Рисунок 6: Информация

2.7 Работа с Midnight Commander

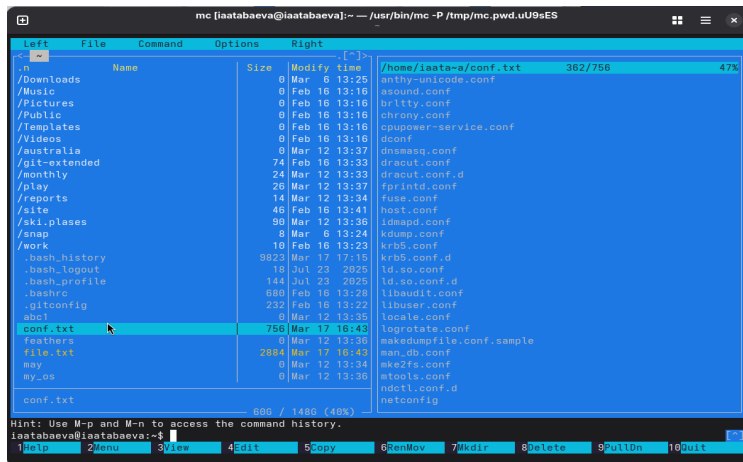


Рисунок 7: Быстрый просмотр

2.8 Работа с Midnight Commander

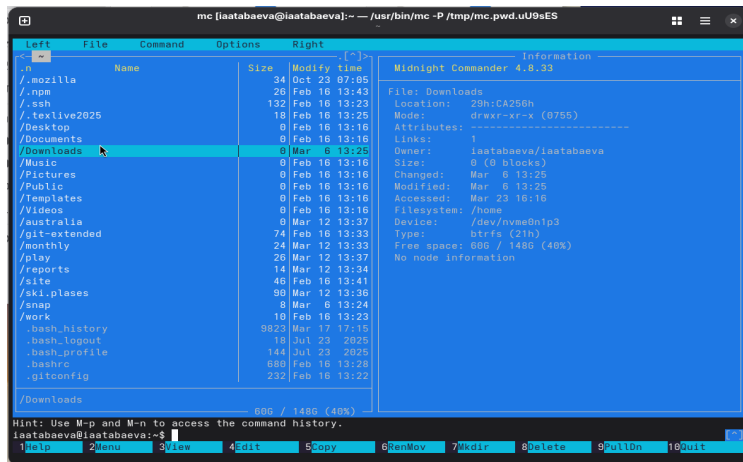


Рисунок 8: Информация

2.9 Работа с Midnight Commander

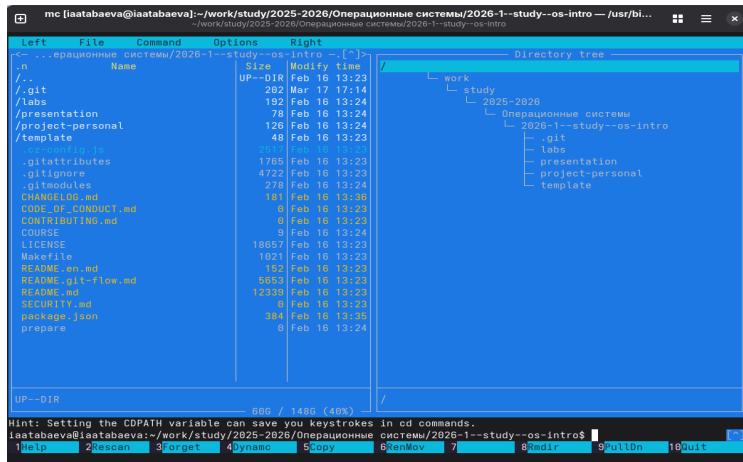


Рисунок 9: Дерево каталогов

2.10 Работа с Midnight Commander



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the current directory is `/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main window displays the contents of a `Makefile` file. The file content includes a shebang, course information, a help target, and a list of submodules. The bottom status bar shows navigation shortcuts: 1Help, 2UnWrap, 3Quit, 4Hex, 5Goto, 6, 7Search, 8Raw, 9Format, 10Quit.

```
mc [iaatabaeva@iaatabaeva]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /usr/bi...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/iaatabaeva/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/Makefile 1021/1021 108%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
    @echo 'Usage:'
    @echo '  make <target>'
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {
FS = ".*?##"; } {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"; } {printf "%s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"; } {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @echo

list: ## List of courses
    @. ./template/config/script/list-courses

prepare: ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule: ## Update submodules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard
origin/$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; git clean -dfx'
```

Рисунок 10: Просмотр содержимого текстового файла

2.11 Работа с Midnight Commander



```
mc [iaatabaeva@iaatabaeva]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /usr/bi...
~/.work/study/2025-2026/Операционные системы/2026-1--study--os-intro
Makefile  [----]  0 L: [ 1+ 0 1/ 28]  *(0 /1021b) 0083 0x053  [*](x)
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {
----->@grep -E '^[###].*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"; {printf "%s\n", $$1, $$2}'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?###"}
----->@echo

list:<-->## List of courses
----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<---->## Update submodules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

2.12 Работа с Midnight Commander

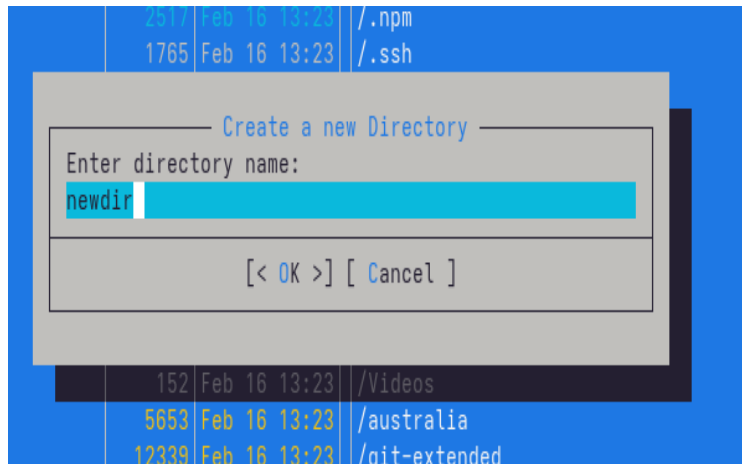


Рисунок 12: Создание каталога

2.13 Работа с Midnight Commander

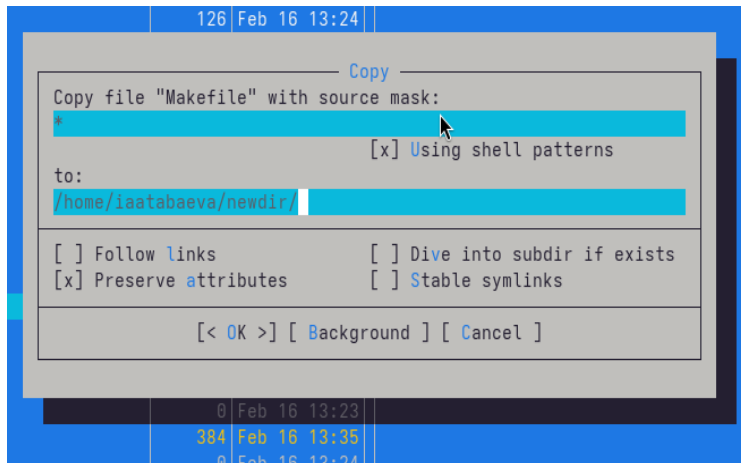


Рисунок 13: Копирование в файлов в созданный каталог

2.14 Работа с Midnight Commander

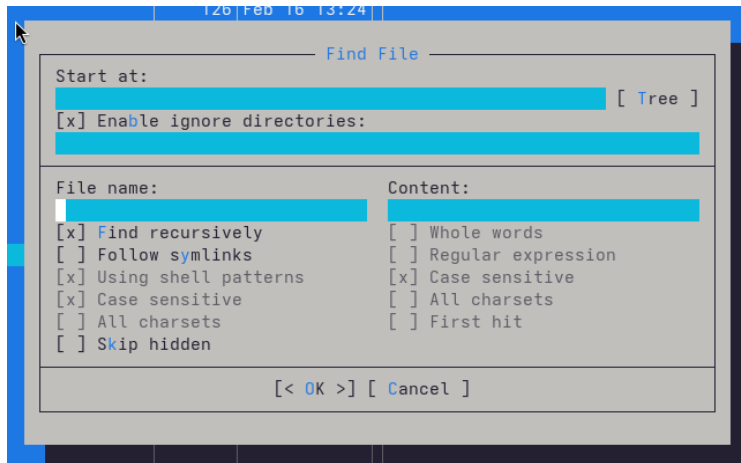


Рисунок 14: Поиск файлов

2.15 Работа с Midnight Commander

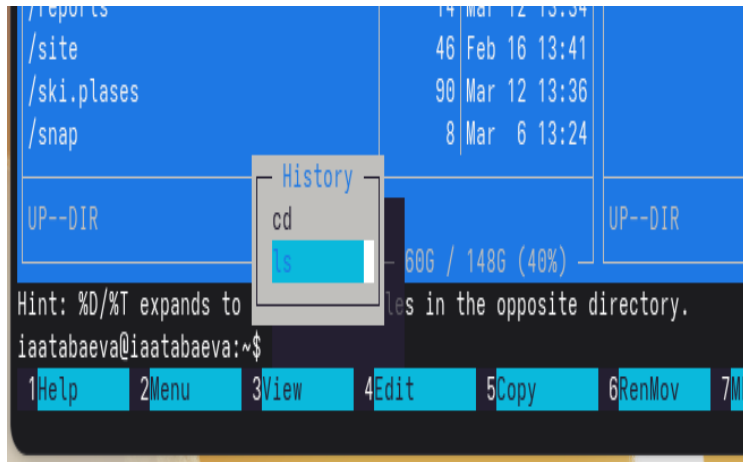


Рисунок 15: История команд

2.16 Работа с Midnight Commander

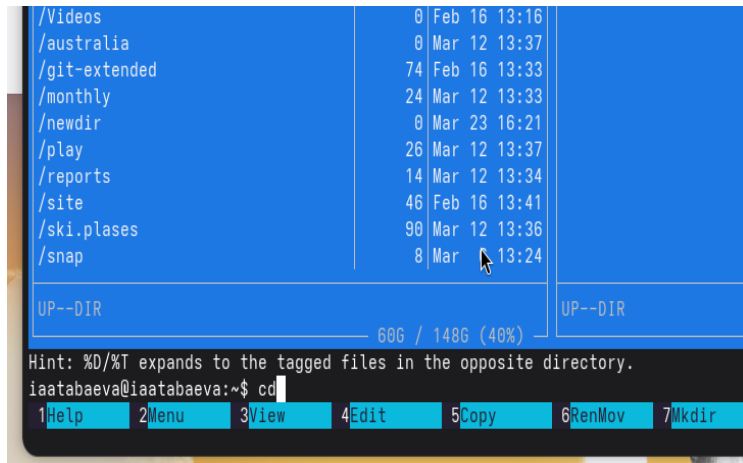
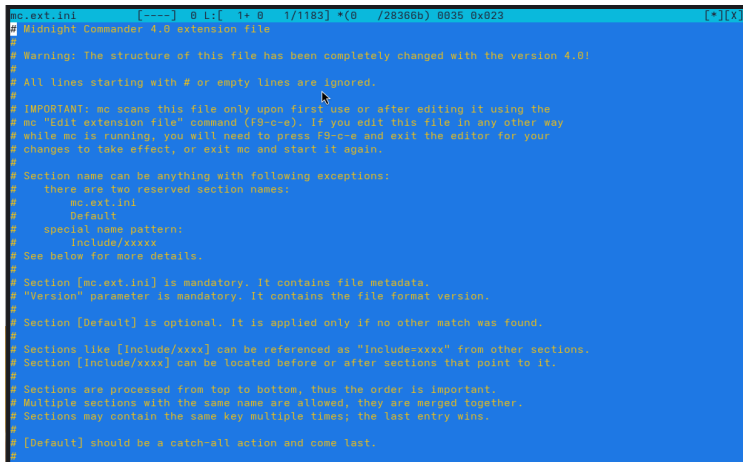


Рисунок 16: Переход в домашний каталог

2.17 Работа с Midnight Commander



```
mc.ext.ini [---] 0 L: 1+ 0 1/1183 *(0 /28366b) 0035 0x023 [*][X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

2.18 Работа с Midnight Commander

```
menu [----] 0 L: [ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#           %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f   quote expanded macro
# %if  ditto
# %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню

2.19 Работа с Midnight Commander

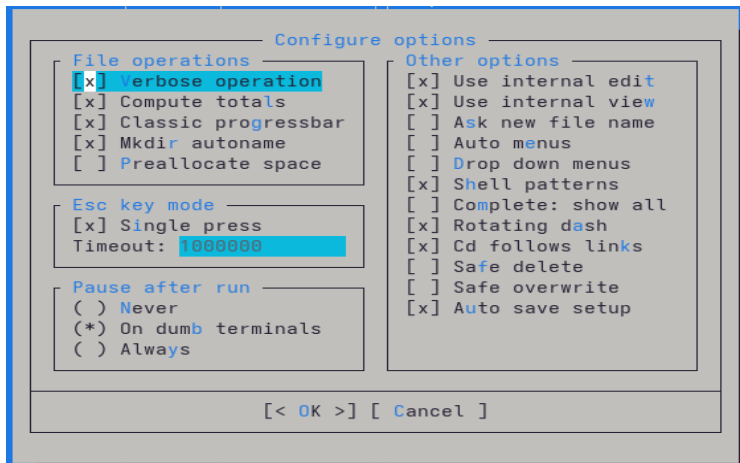


Рисунок 19: Конфигурация

2.20 Работа с Midnight Commander

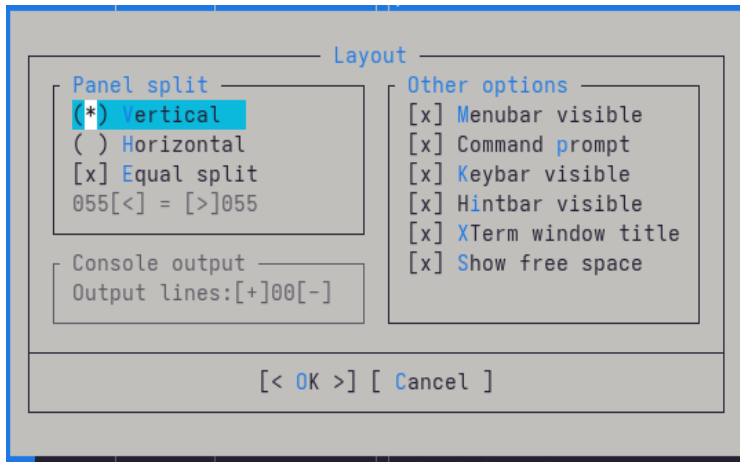


Рисунок 20: Внешний вид

2.21 Работа с Midnight Commander

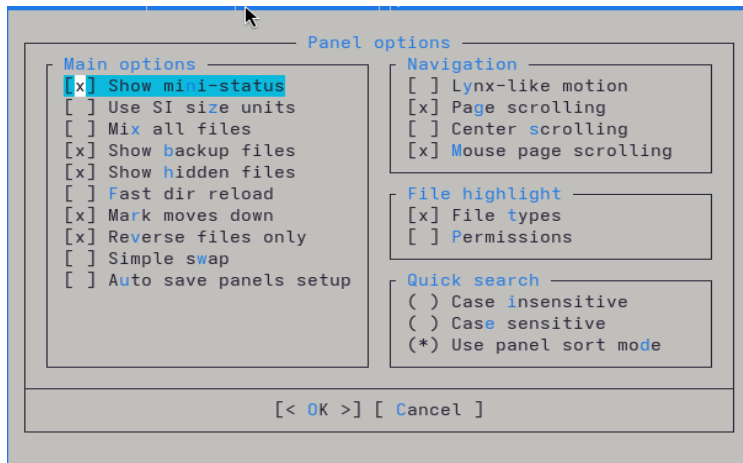


Рисунок 21: Настройки панелей

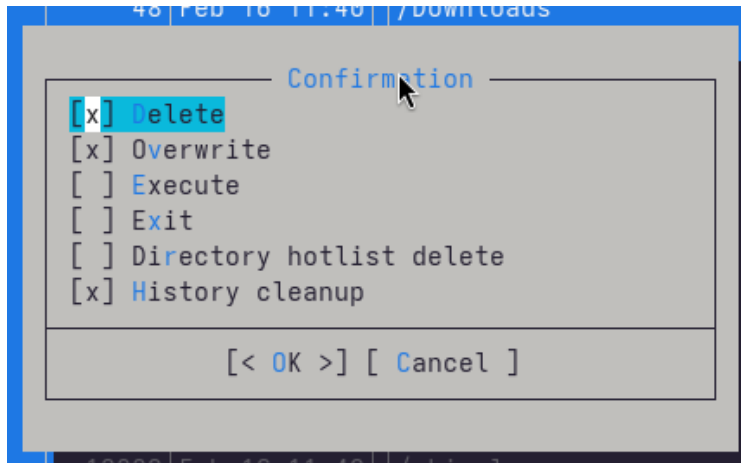


Рисунок 22: Подтверждение

2.23 Работа с Midnight Commander

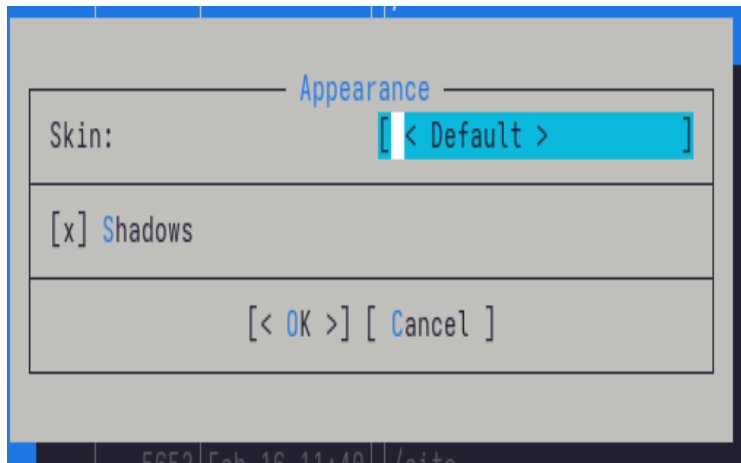


Рисунок 23: Оформление

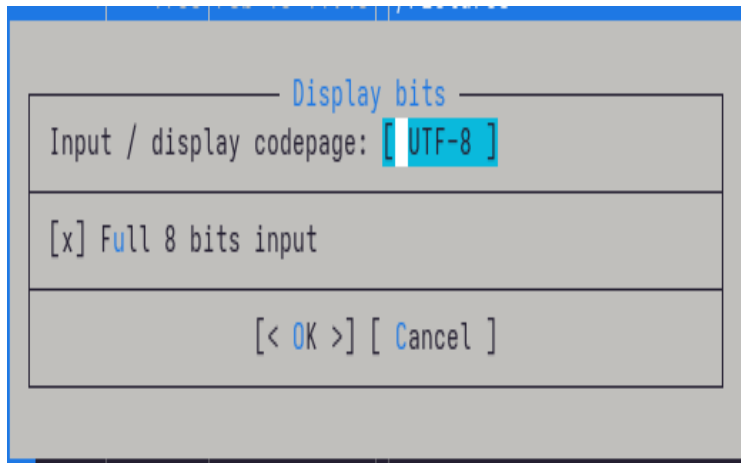


Рисунок 24: Кодировка символов

2.25 Работа с Midnight Commander

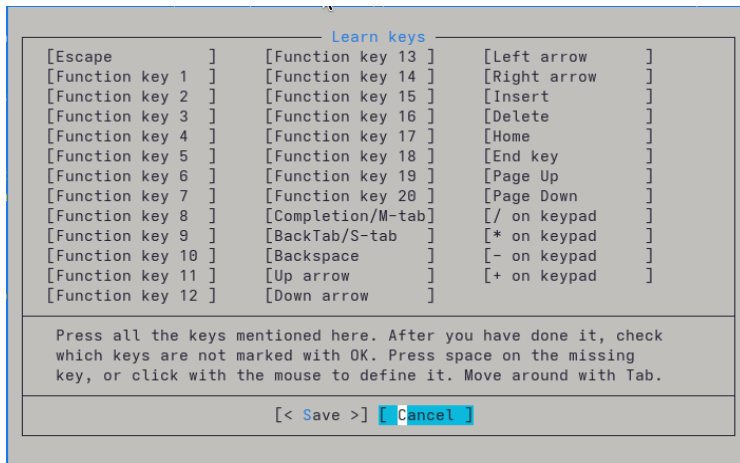
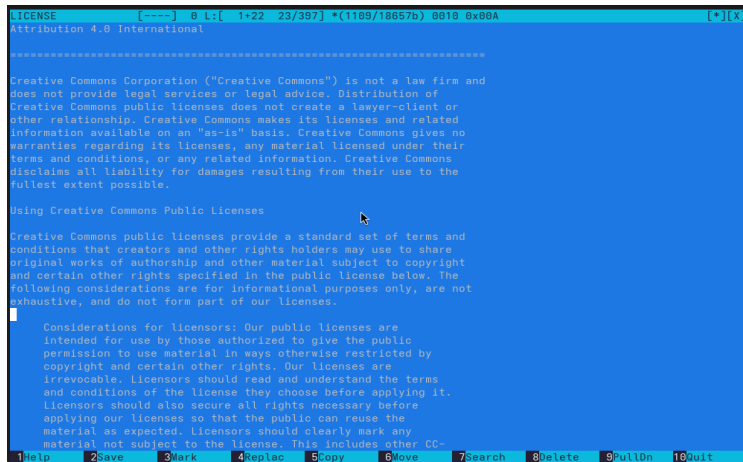


Рисунок 25: Распознавание клавиш

2.26 Работа с редактором Midnight Commander



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
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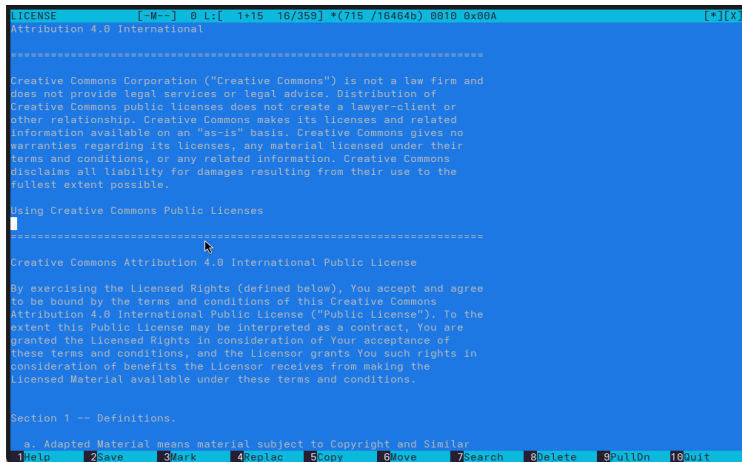
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 26: Файл с текстом

2.27 Работа с редактором Midnight Commander



```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
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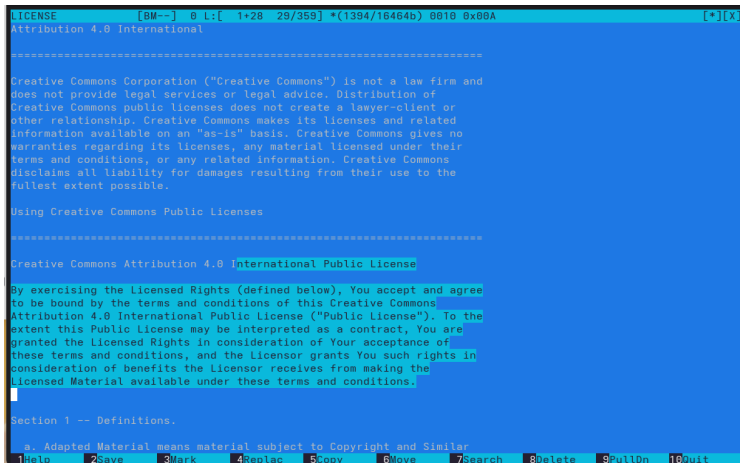
Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом

2.28 Работа с редактором Midnight Commander



```
LICENSE      [BM--]  0 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A      [*][X]
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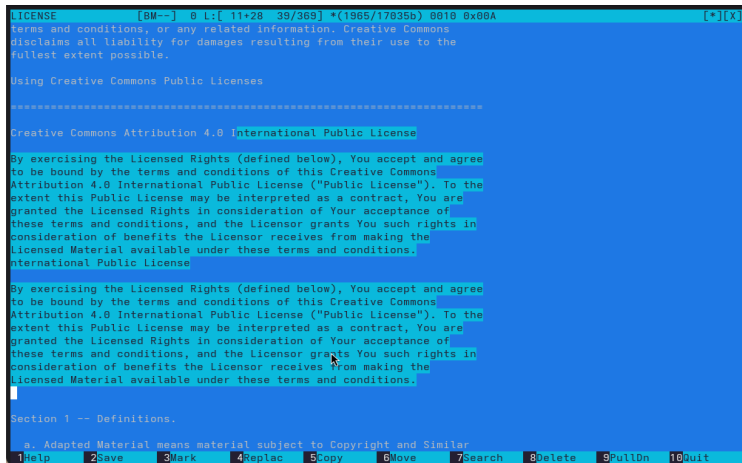
Section 1 -- Definitions.

  a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullOn     10Quit
```

Рисунок 28: Копирование фрагмента

2.29 Работа с редактором Midnight Commander



```
LICENSE      [BM--]  0 L:[ 11+28  39/369] *(1965/17035b) 0010 0x00A      [*][X]
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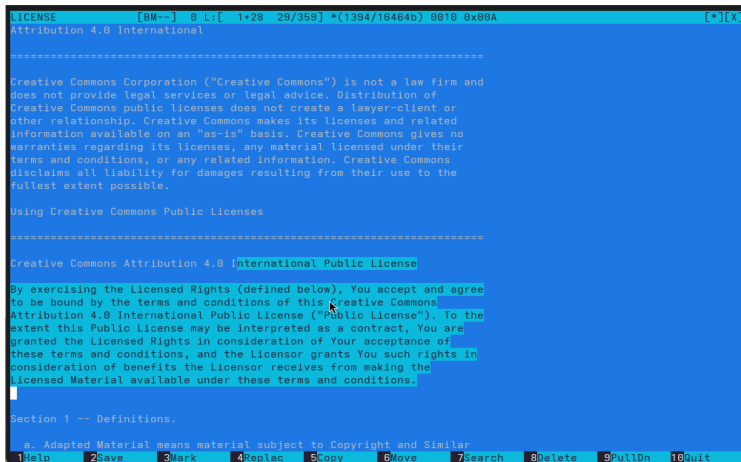
Section 1 -- Definitions.

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1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 29: Сохранение

2.30 Работа с редактором Midnight Commander



```
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 30: Отмена

2.31 Работа с редактором Midnight Commander

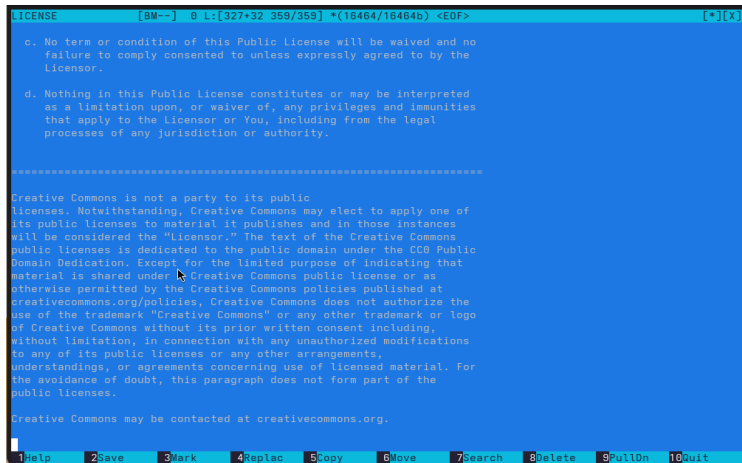
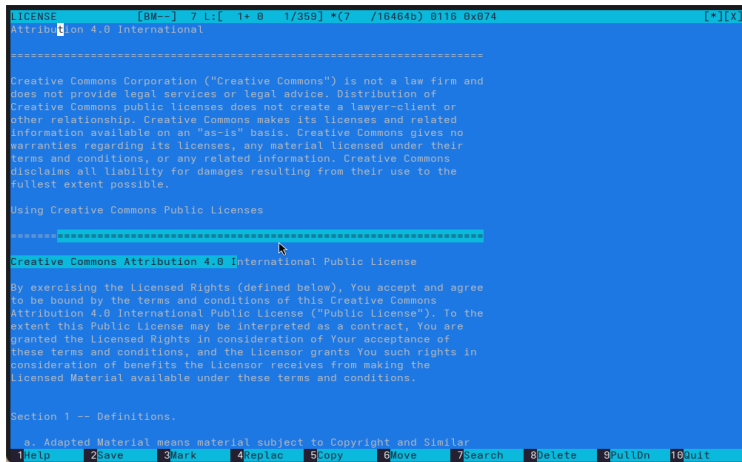


Рисунок 31: Переход в конец файла

2.32 Работа с редактором Midnight Commander



```
LICENSE      [BM--] 7 L:[ 1+ 0 1/359] *(7 /16464b) 0116 0x074      [*][X]
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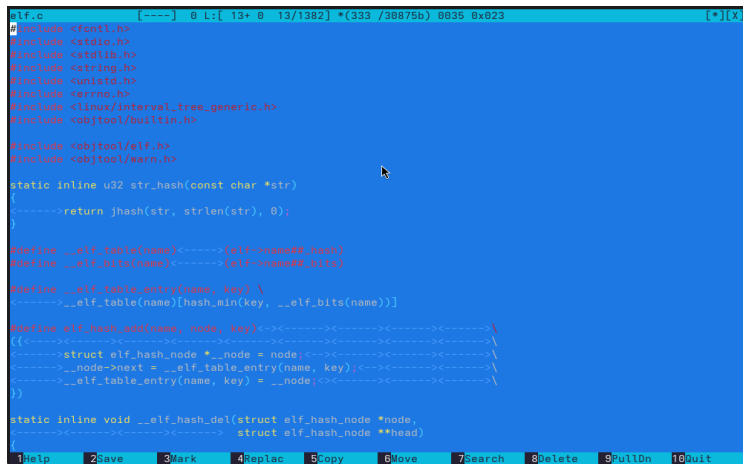
Section 1 -- Definitions.

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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 32: Переход в начало файла

2.33 Работа с редактором Midnight Commander



```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->----->\
    ({----->----->----->----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->----->-----> struct elf_hash_node **head)
```

Рисунок 33: Файл с программой

2.34 Работа с редактором Midnight Commander

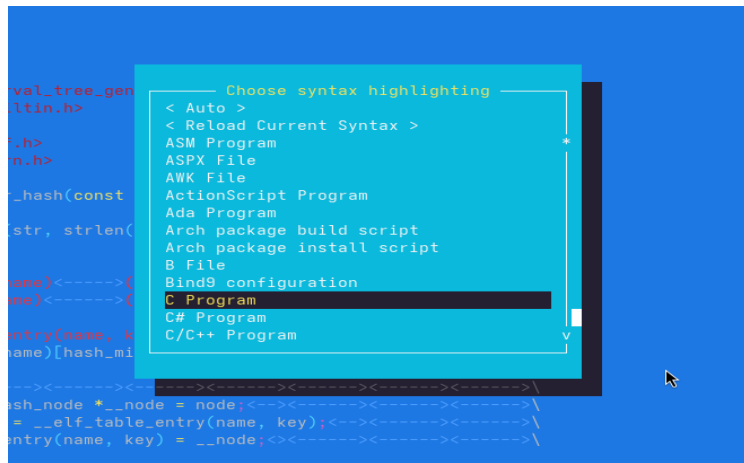


Рисунок 34: Цветовыделение синтаксиса

3. 3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.